Abdullah Mohamed
Ahmed Mostafa
Yassen Mohamed
Youssef Habib
Ziad Tamer

Group: 10116

VerticAlQursayah

10th Grade 1st Semester
(2024-2025)
STEM High School For
boys-6th October

Keywords: vertical-lift-mechanical-bridge-Warren-counterweights

ABSTRACT

Egypt has faced many grand challenges in recent decades. This project aims to solve some of these challenges, including dealing with population growth and urban congestion, and their consequences. Population growth directly contributes to urban congestion as increasing numbers of people migrate to cities in search of opportunities, leading to overcrowded infrastructure, housing shortages, and strained public services. The purpose of this study is present in Giza City near Jazirat Al Qursayah Island, which is an isolated island in the Nile River without any connections with roads to allow vehicle transportation, making the island's construction process more difficult while the Giza City suffers from heavy urban congestion. The selected solution for the present problem is the vertical lift bridge with the warren truss and a lifting mechanism consisting of only one motor and a pulley for mechanical advantage and a weight of 4.5kg The prototype went through many specific tests to ensure that it fulfilled the design requirements, and it successfully carried the required load of 5kg on the midpoint of the longest fixed span, with a deflection of 0.98cm well within acceptable limits. The bridge effectively addressed the intended problem and contributed to tackling Egypt's grand challenge.

INTRODUCTION

Egypt's grand challenges represent severe difficulty to Egypt's development and sustainability plan. These challenges cause serious damage, one of which is population growth which is the increase of civilian percentage in a specific region and population growth has reached about 107,046,149 with an annual growth rate of 1.8%, as shown in **Figure 1**. This problem leads to other major problems as urban congestion which is the increase in population density in metropolitan cities and areas to the point that it exceeds the city's resources causing overcrowding. The problem to be solved is dealing with urban congestion in Giza city and improving connectivity between the city and JaziratAlQursayah island across the Nile River. Solving the problem will connect the island with the road to allow vehicular transportation which will make the construction process on the island easier and more efficient.

While searching for the solution, the first prior solution found is Jacques Chaban, found in France, the tallest vertical-lift bridge in Europe. from its points of strength is that the bridge has a strong structure because it has 4 towers on each corner of the bridge to carry and support the span lifting. But from the its points of weakness is that the mechanism of the bridge takes a long time which disrupts road traffic every time the bridge needs to be raised. Another prior solution is The Erasmus Bridge, in Rotterdam, Netherlands. One of its points of strength is that it was being constructed of light materials as it is made of thin steel allay and pylons are made of reinforced concrete. But from its points of weakness is that the bridge costs more than 165m Euros as it has a large scale.

the selected solution was the vertical lift bridge with the warren truss and a lifting mechanism consisting of only one motor, four towers connected, and a pulley for the mechanical advantage. The Warren truss was selected for its simplicity and suitability for shorter spans, as it eliminates the need for vertical elements to enhance the structure's strength. Constructed following the design requirements.

The design requirements of the bridge are as follows: the scale of the prototype is 1:50, total length of the bridge is between 80cm to 100cm having two lanes for traffic plus two lanes for pedestrians. The width of the traffic lanes and pedestrian lanes are 7cm and 3cm respectively with a 2cm island. it allows the passage of ships with minimum width of 24cm to a maximum width of 28cm and a minimum height of 20cm. it must carry a load of 5kg at the midpoint of the longest span with a deflection not exceeding 5cm. the weight of the prototype doesn't exceed 20kg.

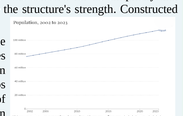


*Figure 1: **population in Egypt from 2002 to 2023***

MATERIALS & METHODS

Table: 1 Shows the Materials used in constructing the prototype

Name	Quantity	Usage	Image
Palm tree wood	30 meters	Sticks of palm tree wood serve as the main construction material for the bridge's body.	
DC geared motor 12V	1 piece	DC geared motor is used for lifting the span.	
Pully	1 pully	The pulley system redirects force.	
Aluminum rods	4 aluminum rods	The aluminum rods are used in the towers in the mechanism.	
MDF spray	200ml spray and 45ml glue	Glue spray is used for fixing the parts together.	
Fishing rope	2 meters	The fishing rope is used in the lifting mechanism of the bridge to lift the movable span.	
nails	35 nails	Added for enhanced strength and durability, securing the deck to the frame.	
Arduino	1 board	Arduino is used for building and controlling electronic projects, including motors.	

•1. The project started with a 3D model **as shown in Figure 2.**

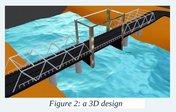


Figure 2: a 3D design

•2. The saw was used to cut the palm tree wood into 35cm,30cm for decks, and 9cm in length, 0.5 in thickness for trusses **as shown in Figure 3**.

•3. The wood had been sanded to be smooth.

•4. The MDF adhesive spray was used to glue the deck together, the trusses together **as shown in Figure 4**.



Figure 3 The process of cutting palm wood tree

• 5. The nails were used to fix the deck more efficiently **as shown in Figure 5**.

•6. The path and the bearings were fixed on each tower in a mechanism using the nails **as shown in Figure 6.**



Figure 4 The process of gluing palm tree

•7. The Arduino was connected to the motor and the pulley.

•8. Next, the motor was put on the left plank and fixed with glue and a rope attached to it to connect the motor with four ropes



Figure 5 The path fixed in wood

Test plan:

• The weight of the prototype, decks, and mechanism were measured.

• The system of the motors was tested to ensure that the motor could lift the bridge.

• The bridge was put on it 5kg to make sure the bridge could bear loads.

• The bridge's deflection was determined under applied loads from 0 to 5 kg to make sure that it didn't cross 5cm in deflection as shown in Figure 7.



Figure 6 The process of fixed decks with nails



Figure 7 Deflection test

RESULTS

Negative results

•The deck pieces were glued using a weak adhesive spray, which caused the pieces to separate. This issue was resolved by replacing the weak spray with a stronger adhesive spray and securing the pieces with nails for better stability.

•Some members of deck were broken because the nails broke the deck as it was hammered into it.

Positive results

The deflection of the bridge was measured by the deflectometer. The bridge had a deflection after putting some loads on the fixed part and the following loads were added in the center of the span 0.5kg,1kg,1.5kg to 5kg, and the average of the results was taken. The deflection increased by increasing the load and the bridge could bear 5kg **as shown in figure 8**. When the load was 5kg, it made a maximum deflection of 1.223 cm and a minimum of 0.9946 cm **as shown in Table 2**, which is less than 5cm and didn't exceed the requirements. The bridge's weight was measured by scale in the physics lab, and the weight of the prototype was 4.5kg and did not exceed 20kg.

Table: 2 Shows the data of the bridge deflection

Criteria	Weight	Average	Maximum	Minimum
Trial 1	0.5kg	0.157cm	0.292cm	0.240cm
Trial 2	1 kg	0.164cm	0.372cm	0.257cm
Trial 3	1.5 kg	0.166cm	0.486cm	0.284cm
Trial 4	2 kg	0.199cm	0.658cm	0.292cm
Trial 5	2.5 kg	0.392cm	0.793cm	0.343cm
Trial 6	3 kg	0.997cm	0.999cm	0.909cm
Trial 7	3.5 kg	1.039cm	1.093cm	0.939cm
Trial 8	4 kg	1.060cm	1.099cm	0.989cm
Trial 9	5 kg	1.102cm	1.223cm	0.995cm

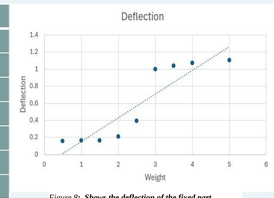


*Figure 8: **Shows the deflection of the fixed part***

ANALYSIS

The project results successfully met the design requirements, incorporating deep scientific foundations addressing the problem of urban conjunction and having to reduce it. these achievements align with the project's goal of addressing Egypt's grand challenges, particularly in improving transportation and increasing urban areas.

Torque and motor efficiency

Torque is critical in designing a vertical lift bridge, as it ensures the motor can lift the bridge deck effectively. Torque measures a motor's ability to rotate and lift a load, depending on its weight and the radius from the axis of rotation. Using a pulley enhances motor performance by creating a mechanical advantage, which reduces the torque required from the motor by effectively changing the load radius. The selected DC motor for this design provides a maximum torque of ≥10 kg cm After converting it to Nm, the torque is equal to 0.981Nm

$$\tau = F \cdot r_{load}$$

Equation 1: formula of torque

Torque (τ) depends on the force applied (F) and the perpendicular distance from the axis of rotation (r_{load}) which is equal to 25 cm.

$$m_{max} = \frac{r_{motor}}{r_{load} \cdot g}$$

Equation 2: formula of maximum mass the motor can handle

Maximum mass is the most mass the motor can handle (m*max*) without the pulley system. The calculation shows that the motor can hold a maximum weight of only 3.924**N**. Without a pulley, the motor must directly handle the full load, which limits the weight it can lift. When adding a pulley, a mechanical advantage is created, and the effective torque increases which raises the weight it can handle. Mechanical advantage (MA) is the factor by which a pulley system multiplies the torque or force. It is the ratio of the load radius (r_{load}) to the pulley radius (r_{pulley}). The pulley reduces the effort required by the motor by increasing the effective torque.

$$\tau_{effective} = MA \cdot \tau_{motor}$$

Equation 3: formula of effective torque after adding the pulley system

In equation 3 Effective torque (τ) is the force (F) exerted by the motor with the pulley This equation helps with calculating the force available to lift the load when the pulley system is introduced. The motor's lifting capacity is determined by the applied force and load radius. after calculating the effective torque which is equal to 9.81Nm and then substituting in the maximum mass equation and then multiplying it by the gravitational force it will be equal to 3.924N without assistance, but when introducing a pulley system it provides a mechanical advantage, increasing its load to weight to 9.81N which is more than enough to hold the weight of the deck.

Truss analysis (method of joints)

The Method of joints is used to solve trusses (solve for forces acting on joints). Also, it specifies the member state if it is in compression or tension, with a negative force value for compression, and positive for tension. Bridge static equilibrium is achieved when the sum of all the horizontal and vertical forces: $\Sigma Fx = 0, \Sigma Fy = 0$

Equation4: The three equilibrium equations

If there isn't horizontal force acting on the truss such as, wind power, the horizontal equilibrium won't be used in analyzing the truss. $\Sigma Fy = 0$ only will be used. As shown in figure 9, consider a Warren truss to have an applied force of **50 N** at the midpoint. To solve for the forces, apply the vertical equilibrium equation (to solve the force on the diagonal):

$$(Reaction\ Force) + (Fdiagonal) \cdot sin(\Theta) = 0$$

Where (θ) refers to the angle between the diagonal and the horizontal member. The force on the diagonal ($Fdiagonal$) will be $28.86N$. The force on the horizontal member according to trigonometry must be $14.43\ N$. By these calculations, the force distribution on all joints in the truss is solved as the force on every member of the truss is equal

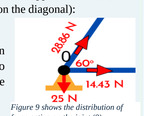


Figure 9 shows the distribution of force acting on the joint (0)

Bending Moment

The bending moment is one of the important calculations required to understand how the bridge deck responds to different loads, especially for a vertical lift bridge design. A bending moment is the measure of the internal forces that oppose the external loads, making it very important to preserve the structural integrity of the deck. The bending moment at a given section is the product of the applied force and its perpendicular distance to the section. In this case, the deck is subjected to a point load applied at its center, which results in the maximum bending moment.

Maximum Bending Moment is calculated along both the length and the width of the deck. For a deck subjected to a point load at its center, calculate the bending moment along each axis length(x) and width(y).

Along the Length of the Deck (M_x)The bending moment at the center of the deck along the length (L) is calculated using the formula:

$$M_x = \frac{P \cdot L}{4}$$

Equation 5: formula of the maximum bending along the x axis(L)

In equation 5 P is the applied load (49.05 N),L is the deck length (0.35 m),b is the deck width (0.22 m).after substituting the maximum bending moment along the length of the deck equal to 4.29Nm

Along the Width of the Deck (M_y)The bending moment along the width (b) is smaller because the load is distributed differently. The formula is:

$$M_y = \frac{P \cdot b}{8}$$

Equation 6: formula of the maximum bending along the y axis (b)

After substituting the maximum bending moment along the width of the deck equal to 1.35Nm In the deck, the resultant bending moment is the sum of the moments that can be expressed as:

$$M_{resultant} = \sqrt{M_x^2 + M_y^2}$$

Equation 7: formula of the maximum bending along the x and y axis

In conclusion, the maximum bending moment for the deck is 4.5 Nm, accounting for both the length and the width of the deck. This value represents the combined effect of the external load and ensures that the design is adequate to withstand the forces without deformation.

Stress and Strain in Bridge

Stress and strain are fundamental concepts in structural engineering, essential for ensuring the safety and durability of materials under loading. In the context of a vertical lift bridge, understanding stress and strain in the deck material is critical to designing a structure that can withstand the applied loads without failure.

Stress (σ) refers to the internal force per unit area within the material, indicating how much load the material can resist. Strain (ϵ) measures the deformation of the material in response to stress. Together, these parameters help engineers assess whether the selected material and design can handle the forces generated during the bridge's operation.

In this case, the deck of the vertical lift bridge is subjected to various loads, and we will calculate both stress and strain based on the maximum bending moment and the material's properties, including Young's modulus, which reflects the material's stiffness. Stress (σ) is calculated using the formula:

$$\sigma = \frac{M_{max}}{Z}$$

Equation 8: formula of stress

In equation 8 σ Stress (Pa), M_{max} Maximum bending moment (Nm), Z Section modulus (m^3). The section modulus Z is calculated using the dimensions of the deck as follows:

$$Z = \frac{b \cdot h^2}{6}$$

Equation 9: formula of section modules

In equation 9 (b) Width of the deck = 0.22 m, and (h) the Thickness of the deck = 0.011m. After substituting the section modulus equal to $4.433 \times 10^{-6} m^3$.Using the maximum bending moment of 4.5 Nm, substitute in the equation 9 which will be equal to 1,016.85kPa.Strain (ϵ) is calculated using the formula:

$$\epsilon = \frac{\sigma}{E}$$

Equation 10: formula of strain

ϵ: Strain (dimensionless), σ: Stress (Pa),the materials were chosen for its young's modules where the highest young's modules between the midribs of the palm tree are siwi palm tree. after substituting in equation 10 strain equal to 1.37×10^{-4}

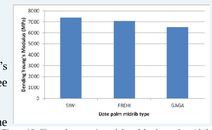


Figure 10: Shows the young's modules of the date palm midrib

The Young's modulus of the material is given as E=7395 Mpa integrity of the design.

The **stress** in the deck under the given load is 1,016.85 kPa, The **strain** is 1.37×10^{-4}The **Young's Modulus** of the material is 7395 MPa. These values ensure that the material used for the bridge deck can withstand the applied loads without experiencing permanent deformation, contributing to the overall structural

Table: 3 Shows the LOs which was used in the project

learning outcome (LO)	Concept	usage
PH.1.01	Measurement errors	used while measuring the dimensions of the bridge to improve the measurement accuracy and precision.
MA.1.01	Sine and cosine law	used for calculating unknown sides and angles between truss members.
MA.1.02	histograms	Used in visualizing key performance metrics like material properties, load capacity, and deflection behavior. identifying variations, optimizing design, and analyzing errors.
MA.1.04	similarity	Used in scaling objects in the bridge, analyzing proportions, and solving problems.
PH.1.02	System of forces	used in analyzing and balancing the forces acting on the structure, calculating the resultant forces from loads, such as the load weight and the bridge itself

CONCLUSIONS

Based on the previous research on the grand challenges that Egypt faces, especially population growth, dealing with urban congestion, and improving connectivity across the Nile River water streams, it was found that the best solution for this problem is constructing a vertical-lift bridge to achieve the best efficiency. The bridge's mechanism consists of four connected towers, a single high-torque motor, a pulley, and a wire connected to the four vertexes of the movable span. After the project was tested, the performance of the bridge yielded promising results, showing how it balances the applied weight, moves vertically in a smooth way, and handles applied forces. Constructing and testing the structure of the bridge provided key insights into how the materials interact, the effectiveness of the lifting mechanism, and the accurate distribution of forces. This prototype not only confirmed the design's ideas but also revealed areas that could be improved for a real-life one. Overall, the prototype is an important step toward creating a safe, reliable, and efficient bridge that could solve the problems concerned with urban congestion, population growth, and improving connectivity across the Nile River water streams between Giza and Jazirat al Qursaya.

RECOMENDATION

Real-life recommendation

When the bridge is constructed in real life connecting Jazirat Al Qursayah island and the city of Giza, it will solve the problem of no connection between the island and the road allowing vehicular transportation which will make the construction process on the island easier and more efficient. From the crucial points that needs to be focused on is the silty soil on which the bridge will be constructed, which can support the building of structures like a house, not a bridge that has frequent and different loads, as it needs to be more stable. stability can be achieved by adding natural pozzolan and lime to the soil the compressive will increase strength significantly, but adding them together is more effective. So, the compressive strength of a treatment having 3% lime and 15% pozzolan was found to be 16 times more effective. The use of different materials like aluminum-steel alloy as it is characterized with light weight and its durability, and the use of counterweights is very important where in real life the motor wouldn't handle the dynamic load. Furthermore, the use of Reinforced Concrete is used in towers and foundation of vertical lift bridge it has high durability and can be easily shaped It combines concrete's high compressive strength with steel's excellent tensile properties.



Figure 11: Shows the location

Types of supports

Changing of the supports to one movable support and one fixed instead of two fixed supports in the prototype which would improve the efficiency and structural integrity of the bridge. The fixed support would prevent the bridge from rotational displacement. On the other hand, the movable support would allow thermal expansion and contraction and, at the same time, permit small displacements due to dynamic loads like traffic. Such a design would allow for more even force distribution across the bridge to reduce concentrations and therefore slow down structural deterioration like earthquakes. By allowing adaptation to all environmental and changing load conditions, the current design would greatly help increase durability, as well as effectiveness through its operation.

105 Epoxy Resin / 205 Fast Hardener



*Figure 12:*105 Epoxy Resin / 205 Fast Hardener

105/205 Epoxy is used for general coating and bonding applications. It forms a high-strength, moisture-resistant solid with excellent bonding and barrier coating properties. It can be used in the prototype of the bridge to coat the deck and truss, but it isn't available in Egypt and it needs a higher budget to be purchased. It will wet out and bond to wood fiber, a variety of metals, and other composite materials. It can be thickened with fillers to bridge gaps and fill voids and can be sanded and shaped. It can improve tensile strength, tensile modulus, hardness, and compressive yield, which will improve the durability, strength, and functionality of the bridge, making it resist more heavy loads.

Addition of motors

Designing a lift bridge with four motors instead of one will result in better balance, efficiency, and reliability. Distribution of this load across four motors puts less stress on each, which increases the life of each motor and reduces the risk associated with mechanical failure. This structure also allows for smooth lifting to result in even movement with little stress on the bridge structure. In the case of a motor failure, other operating motors can provide limited capability, which provides an extra degree of safety. Besides, multiple motors contribute to better scalability, allowing the system to operate with heavier loads and more stability. This will also offer a larger scale for the prototype where the use of one motor limits the weight and length of span it could handle on the other way making four motors offers a larger scale and weight it handles in the prototype.

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ACKNOWLEDGMENTS

First of all, We are indebted to all those people who assisted us through this journey. Thank's alot to Mr. Mohamed Bekheet (Capstone General), Mrs. Dalia Hamed (Capstone Leader), The valuable time, guidance, knowledge, and encouragement of Mohamed Ali, (Capstone Teacher), and Dr. Metwally Abu Hamad, (Engineering Doctor at Cairo University), merit our thanks. Without their valuable assistance, this project would not have seen the light of day. We are indeed very thankful for all of the support they have given us.